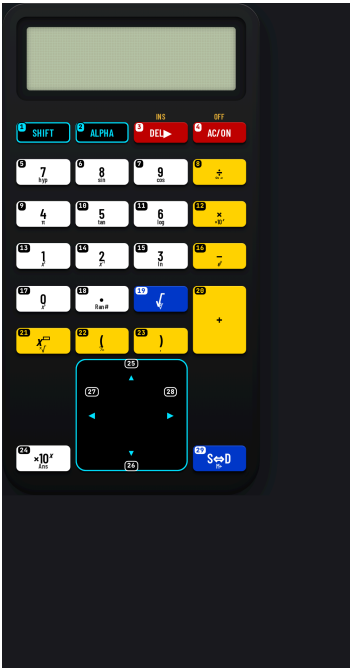
# U-45 CALCULATORS

## User Manual

### Calculator #41 Compact Pocket Scientific (black, green DEL/AC)

Scientific – compact pocket




Complete kit: functions · numbered key map · worked examples · tricks & hacks · shortcuts · common mistakes · practice drills

# About this manual

This manual belongs to the **U-45 CALCULATORS** app and describes calculator **#41 — Compact Pocket Scientific (black, green DEL/AC)** exactly as it behaves in the app. Every worked example and every drill answer in this book was run on the app's own keys by an automated checker before printing; only results that matched were kept.

## How key presses are written

Keys are written as a chip showing the **key number** and its face, for example [#1 SHIFT]. Numbers follow the numbering printed on the keys (switch it on with "# On"). A shifted function is written as two chips: the shift key, then the key that carries the function.

## Entry style of this model

algebraic (type the expression as written, then = / EXE).

## Chapters

1. Getting started
2. Features
3. Numbered key map
4. Key-by-key reference
5. Worked examples
6. Tricks & hacks
7. Keyboard shortcuts
8. Common mistakes
9. Practice drills
10. Appendix A — Drill answers
11. Appendix B — Function finder (A–Z)

12. Appendix C — Glossary

13. Appendix D — Model notes & limits

14. Appendix E — Quick-reference card

# 1 · Getting started

Open U-45 CALCULATORS, choose **#41** from the home screen (or the jump list at the top), and the calculator appears with its numbered, colour-coded keys. Click or tap keys, or type on a keyboard.

### Numbers on every key

Every key carries a small number in the same reading order as this manual (left → right, top → bottom; the navigation pad counts ▲ ▼ ◀ ▶ then its centre). Use "# On / # Off" in the bar above the calculator to show or hide them — the setting is remembered.

### High-contrast colours

Keys are coloured by job: white = digits, yellow = operators/execute, blue = functions, green = memory/variables, red = clear/delete/power, black with cyan = shift/modes/menus/navigation. Press "Classic" to return to the original model colours and "Hi-Contrast" to come back.

### Three typefaces

Display text uses the U45 Display face, key legends use U45 Keys (a condensed face that stays readable on small keys) and key numbers / helper text use U45 Mono. All three are bundled inside the app, so they work offline.

### Copy the display

The Copy button copies exactly what the display shows (indicators that are switched off are skipped) so you can paste a result into a message or spreadsheet.

### Fit or 1:1

Fit scales the calculator to your screen; 1:1 shows it at its design size. On a phone, Fit is the comfortable choice.

### Reset

Reset rebuilds this calculator from scratch: display, memory, stack and modes return to their start-up state. Use it before each worked example in this manual.

### Keyboard

On a computer you can type: digits, + − * / ^ ( ), Enter or = to execute, Backspace to delete, Esc to clear, arrows to move. Letters reach common functions (see the Keyboard shortcuts chapter).

### Tooltips

Hover over (or long-press) a key to see "#number name".

### Tools rail (v1.4)

A cyan icon rail on the right of every calculator opens 15 panels: Currency (166 currencies, rates auto-updated from open.er-api.com or Frankfurter when online and cached with a date; the bundled dated snapshot is used offline), Units (exact SI factors), Health (published formulas, not medical advice), Simple, Complex, Loans, Construction, Mechanical and Engineering (30+ numbered tools each), Calendar generator, Customize (26 options, saved per calculator), Settings (31 options), History, Hide and View. On narrow phones the rail shows icons only.

### History (v1.4)

Every calculation is logged per calculator with a date and time. Open History on the rail to search, re-use (types the input back into the calculator), copy, delete one, select several and delete, or delete all (asks for confirmation). History is stored encrypted on this device (AES-GCM); exporting it needs the passcode.

### Camera Solve (v1.4)

The green CS button between Restore and Social reads a problem from the camera, a photo, a PDF or a text file. Recognition (Tesseract OCR, English) runs offline on the device; an online language download is only used if you switch it on in Settings. Check and edit the recognised text, then Solve shows the steps offline. OCR is not available when the app is opened as a plain file:// page in a browser.

### Layouts and view modes (v1.4)

The layout selector between Fit and # offers Auto plus 12 layouts: 3D classic, 3D high-contrast, Classic, Pocket, Compact, Expanded, Desktop wide, Split, One-hand left, One-hand right, Large print and Minimal. Auto picks one for your screen and your choice is remembered per calculator. The View panel hides the top bars, the tool bar, the rail or everything (full screen); the round yellow ⤢ button at bottom-left (or Esc) always restores the normal view.

### Hide items (v1.4)

The Hide panel lists every button, rail icon and section of this calculator as tick boxes. Hiding is instant; unhiding asks for the app passcode.

### Image protection (v1.4)

Right-click, drag and long-press saving of images is blocked inside the app. Saving or sharing an image export (PNG, JPG, QR, calendar), printing a calendar, and downloading or sharing the app itself ask for the passcode, and exported images carry a U-45 watermark. Honest note: this cannot stop a screenshot or screen recording made by the operating system or another device.

### Works offline

U-45 CALCULATORS is an offline app: after the first load everything — including fonts — comes from the device.

## Colour groups on this calculator

| Colour | Group | Keys on this model |
|---|---|---|
| num | Digits & number entry | #5, #6, #7, #9, #10, #11, #13, #14, #15, #17, #18, #24 |
| op | Operators & execute | #8, #12, #16, #20, #21, #22, #23 |
| fn | Functions | #19, #29 |
| mem | Memory, variables & registers | — |
| clr | Clear, delete & power | #3, #4 |
| mod | Shift, modes, menus & navigation | #1, #2, #25, #26, #27, #28 |

# 2 · Features

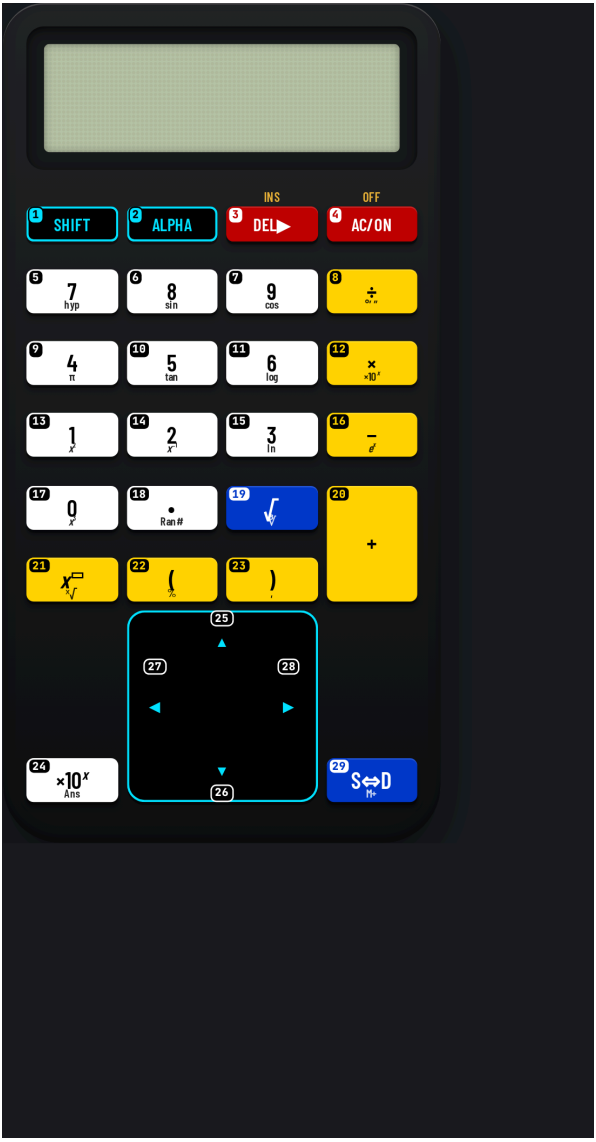
**50 features: 28 live on the keys, 9 app/UI features, 13 planned (documented but not yet working — they are clearly marked and never used in examples).**

1. LIVE **Pocket two-line display**
2. LIVE **S⇔D acts as "=" and fraction/decimal toggle**
3. LIVE **SHIFT functions printed under each key**
4. LIVE **Any-index root and $x^3$**
5. LIVE **M+ independent memory**
6. LIVE **S⇔D / F⇔D toggle between exact (fraction, √, π) and decimal answers**
7. LIVE **Square root √**
8. LIVE **Cube root $\sqrt[3]{}$**
9. LIVE **Any-index root $\sqrt[x]{y}$**
10. LIVE **Square $x^2$**
11. LIVE **Cube $x^3$**
12. LIVE **Reciprocal $x^{-1}$ (1/x)**
13. LIVE **Power $x^y$ / ^ with right-to-left chaining (2^3^2 = 512)**
14. LIVE **Common logarithm $\log_{10}$**
15. LIVE **Natural logarithm ln**
16. LIVE **Exponential $e^x$**
17. LIVE **Trigonometry sin / cos / tan honouring the current angle unit**
18. LIVE **Hyperbolic sinh / cosh / tanh and their inverses via hyp**
19. LIVE **Sexagesimal ° ′ ″ entry and decimal ↔ DMS conversion**
20. LIVE **Random number Ran# (0 ≤ r < 1)**

21. LIVE **Percent key (x %, add-on and discount chains)**
22. LIVE **π constant to full internal precision**
23. LIVE **Scientific-notation entry ×10ˣ / EXP / EEX**
24. LIVE **Ans recall of the last result; operators auto-insert Ans**
25. LIVE **Independent memory M+**
26. LIVE **Backspace / DEL editing of the current entry**
27. LIVE **INS insert mode for editing in the middle of an expression**
28. LIVE **Replay ▲ ▼ — scroll back to previous calculations and re-edit them**
29. APP/UI **Every key numbered 1…N in reading order — matches the key map in the user manual**
30. APP/UI **High-contrast 6-colour key groups (digits, operators, functions, memory, clear, modes) with a "Classic" toggle**
31. APP/UI **Three bundled offline fonts: Oxanium (display/titles), Barlow Condensed (key faces), JetBrains Mono (key numbers & helpers)**
32. APP/UI **Show / hide key numbers with one button (remembered between sessions)**
33. APP/UI **Copy the current display text to the clipboard**
34. APP/UI **Physical-keyboard shortcuts: digits, + – * / ^ ( ) . Enter Backspace Esc arrows, s c t l n r ! % e p q i**
35. APP/UI **Fit / 1:1 zoom that never pushes keys off a phone screen**
36. APP/UI **Reference photo viewer and tool-tip name on every key**
37. APP/UI **Reset = fresh power-on state; works fully offline (PWA cache / Android APK)**
38. PLANNED **Statistics mode**
39. PLANNED **Base-N mode**

40. PLANNED **Equation solver**
41. PLANNED **Table mode**
42. PLANNED **Constants**
43. PLANNED **Unit conversions**
44. PLANNED **Replay history**
45. PLANNED **Complex numbers**
46. PLANNED **Exam mode (locks memory & shows indicator)**
47. PLANNED **Answer history export to text**
48. PLANNED **Speech read-out of results**
49. PLANNED **Colour-blind safe key theme**
50. PLANNED **Custom shortcut keys**

# 3 · Numbered key map




Calculator #41 with key numbers shown (Hi-Contrast colours).

## Key table

**“Shift” = 2nd / f / SHIFT layer; “Alpha” = ALPHA layer; “3rd” = 3rd / g layer. A dot means the layer is empty on that key.**

| # | Face | Shift | Alpha | 3rd | Group |
|---|---|---|---|---|---|
| 1 | SHIFT | · | · | · | mod |
| 2 | ALPHA | · | · | · | mod |
| 3 | DEL▶ | INS | · | · | clr |
| 4 | AC/ON | OFF | · | · | clr |
| 5 | 7 | · | · | · | num |
| 6 | 8 | · | · | · | num |
| 7 | 9 | · | · | · | num |
| 8 | ÷ | · | · | · | op |
| 9 | 4 | · | · | · | num |
| 10 | 5 | · | · | · | num |
| 11 | 6 | · | · | · | num |
| 12 | × | · | · | · | op |
| 13 | 1 | · | · | · | num |
| 14 | 2 | · | · | · | num |
| 15 | 3 | · | · | · | num |
| 16 | − | · | · | · | op |
| 17 | 0 | · | · | · | num |
| 18 | • | · | · | · | num |
| 19 | √ | · | · | · | fn |
| 20 | + | · | · | · | op |
| 21 | x▭ | · | · | · | op |
| 22 | ( | · | · | · | op |
| 23 | ) | · | · | · | op |

| # | Face | Shift | Alpha | 3rd | Group |
|---|---|---|---|---|---|
| 24 | ×10x | · | · | · | num |
| 25 | ▲ | · | · | · | mod |
| 26 | ▼ | · | · | · | mod |
| 27 | ◀ | · | · | · | mod |
| 28 | ▶ | · | · | · | mod |
| 29 | S⇔D | · | · | · | fn |

# 4 · Key-by-key reference

Every key on calculator #41, in number order, with every function it carries in this app.

### #1 SHIFT `Shift, modes, menus & navigation`

**Primary:** SHIFT / 2nd — the next key uses its upper (shift-coloured) function

### #2 ALPHA `Shift, modes, menus & navigation`

**Primary:** ALPHA — the next key types its letter/variable (red/green label)

### #3 DEL▶ `Clear, delete & power`

**Primary:** DEL — delete the character at the cursor

**shifted (2nd / f / SHIFT layer) (press #1 first):** INS — INS — insert mode at the cursor

**Keyboard:** `Backspace` `Delete`

### #4 AC/ON `Clear, delete & power`

**Primary:** ON / AC — power on, or clear all

**shifted (2nd / f / SHIFT layer) (press #1 first):** OFF — OFF — power off (state is lost; press ON to restart)

### #5 7 Digits & number entry

**Primary:** Digit 7

**shifted (2nd / f / SHIFT layer) (press #1 first):** — Hyperbolic prefix — the next sin/cos/tan becomes sinh/cosh/tanh

**Printed under the key:** hyp

**Keyboard:** 7

### #6 8 Digits & number entry

**Primary:** Digit 8

**shifted (2nd / f / SHIFT layer) (press #1 first):** — Sine (uses the current angle unit)

**Printed under the key:** sin

**Keyboard:** 8

### #7 9 Digits & number entry

**Primary:** Digit 9

**shifted (2nd / f / SHIFT layer) (press #1 first):** — Cosine (current angle unit)

**Printed under the key:** cos

**Keyboard:** 9

**#8** ÷ `Operators & execute`

**Primary:** Division ÷

**shifted (2nd / f / SHIFT layer) (press #1 first):** — Degrees-minutes-seconds entry/display °′ ″

**Printed under the key:** °′ ″

**Keyboard:** /

**#9** 4 `Digits & number entry`

**Primary:** Digit 4

**shifted (2nd / f / SHIFT layer) (press #1 first):** — π (3.14159265…) as an exact constant

**Printed under the key:** π

**Keyboard:** 4

**#10** 5 `Digits & number entry`

**Primary:** Digit 5

**shifted (2nd / f / SHIFT layer) (press #1 first):** — Tangent (current angle unit)

**Printed under the key:** tan

**Keyboard:** 5

## #11 6 Digits & number entry

**Primary:** Digit 6

**shifted (2nd / f / SHIFT layer) (press #1 first):** — Common logarithm $\log_{10}$

**Printed under the key:** log

**Keyboard:** 6

## #12 × Operators & execute

**Primary:** Multiplication ×

**shifted (2nd / f / SHIFT layer) (press #1 first):** — $\times 10^x$ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 $\times 10^x$ 23

**Printed under the key:** ×10x

**Keyboard:** * x

## #13 1 Digits & number entry

**Primary:** Digit 1

**shifted (2nd / f / SHIFT layer) (press #1 first):** — Square $x^2$

**Printed under the key:** $x^2$

**Keyboard:** 1

## #14 2 Digits & number entry

**Primary:** Digit 2

**shifted (2nd / f / SHIFT layer) (press #1 first):** — Reciprocal $x^{-1}$ (1/x)

**Printed under the key:** $x^{-1}$

**Keyboard:** 2

**#15** **3** `Digits & number entry`

**Primary:** Digit 3

**shifted (2nd / f / SHIFT layer) (press #1 first):** — Natural logarithm ln

**Printed under the key:** ln

**Keyboard:** 3

**#16** **–** `Operators & execute`

**Primary:** Subtraction –

**shifted (2nd / f / SHIFT layer) (press #1 first):** — $e^x$ — exponential

**Printed under the key:** ex

**Keyboard:** -

**#17** **0** `Digits & number entry`

**Primary:** Digit 0

**shifted (2nd / f / SHIFT layer) (press #1 first):** — Cube $x^3$

**Printed under the key:** $x^3$

**Keyboard:** 0

**#18** **•** `Digits & number entry`

**Primary:** Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored

**shifted (2nd / f / SHIFT layer) (press #1 first):** — Random number 0–1

**Printed under the key:** Ran#

**Keyboard:** . ,

## #19 √ Functions

**Primary:** Square root √

**shifted (2nd / f / SHIFT layer) (press #1 first):** — Cube root ∛

**Printed under the key:** ∛

**Keyboard:** r

## #20 + Operators & execute

**Primary:** Addition +

**Keyboard:** +

## #21 x▭ Operators & execute

**Primary:** Power $x^y$ / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)

**shifted (2nd / f / SHIFT layer) (press #1 first):** — n-th root $\sqrt[x]{y}$

**Printed under the key:** $\sqrt[x]{}$

**Keyboard:** ^

## #22 ( Operators & execute

**Primary:** Opening bracket (

**shifted (2nd / f / SHIFT layer) (press #1 first):** — Percent %

**Printed under the key:** %

**Keyboard:** (

## #23 ) Operators & execute

**Primary:** Closing bracket ). Closing brackets at the end of a line may be omitted

**shifted (2nd / f / SHIFT layer) (press #1 first):** — Comma / argument separator (between function arguments, list items or coordinates)

**Printed under the key:** ,

**Keyboard:** )

## #24 ×10x Digits & number entry

**Primary:** $\times 10^{x}$ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 $\times 10^{x}$ 23

**shifted (2nd / f / SHIFT layer) (press #1 first):** — Ans — recall the last answer into the current expression

**Printed under the key:** Ans

**Keyboard:** e

## #25 ▲ Shift, modes, menus & navigation

**Primary:** Cursor ▲ — up / previous history line / previous item

## #26 ▼ Shift, modes, menus & navigation

**Primary:** Cursor ▼ — down / next history line / next item

## #27 ◀ Shift, modes, menus & navigation

**Primary:** Cursor ◀ — move left in the line

#28 ▶ Shift, modes, menus & navigation

**Primary:** Cursor ▶ — move right in the line

#29 S⇔D Functions

**Primary:** S⇔D — toggle the result between exact (fraction, √, π) and decimal

**shifted (2nd / f / SHIFT layer) (press #1 first):** — M+ — add the value to independent memory M

**Printed under the key:** M+

# 5 · Worked examples

**19 examples, each run on this calculator's keys by the U-45 checker. Follow the chips in order.**

### Example 5.1 — Order of operations

**Calculate 12 + 3 × 4. Press Reset first.**

#13 1 → #14 2 → #20 + → #15 3 → #12 × → #9 4 → #29 S⇔D

**Display shows** 24 **— expected 24. ✔ verified on the app.**

*Multiplication is done before addition, so the calculator holds the + until the × part is finished.*

### Example 5.2 — Brackets first

**Calculate (12 + 8) ÷ 4. Press Reset first.**

#22 ( → #13 1 → #14 2 → #20 + → #6 8 → #23 ) → #8 ÷ → #9 4 → #29 S⇔D

**Display shows** 5 **— expected 5. ✔ verified on the app.**

*Brackets force the sum to be worked out before the division.*

### Example 5.3 — Two brackets multiplied

**Calculate (3 + 4) × (9 − 2). Press Reset first.**

#22 ( → #15 3 → #20 + → #9 4 → #23 ) → #12 × → #22 ( → #7 9 → #16 − → #14 2 → #23 ) → #29 S⇔D

**Display shows** 49 **— expected 49. ✔ verified on the app.**

*Both brackets are evaluated first, then multiplied.*

### Example 5.4 — Squares

**Calculate $12^2$. Press Reset first.**

#13 1 → #14 2 → #1 SHIFT → #13 → #29 S⇔D

**Display shows** 144 **— expected 144. ✔ verified on the app.**

*$x^2$ squares the value — quicker and safer than typing the number twice.*

### Example 5.5 — Square roots

**Calculate √144. Press Reset first.**

#19 √ → #13 1 → #9 4 → #9 4 → #29 S⇔D

**Display shows** 12 **— expected 12. ✔ verified on the app.**

*The square-root key returns the principal (positive) root.*

### Example 5.6 — Powers

Calculate **2^10**. Press **Reset** first.

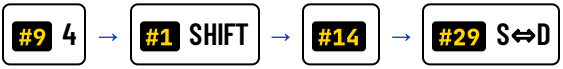



Display shows `1024` — expected 1024. ✔ verified on the app.

*The power key raises the first number to the second.*

### Example 5.7 — Reciprocals

Calculate **1 ÷ 4 ($x^{-1}$)**. Press **Reset** first.




Display shows `0.25` — expected 0.25. ✔ verified on the app.

*$x^{-1}$ (1/x) gives the reciprocal — handy for resistors in parallel or rates.*

### Example 5.8 — Trigonometry (degrees)

Calculate **tan 45° (Deg mode)**. Press **Reset** first.




Display shows `1` — expected 1. ✔ verified on the app.

*Trig functions use the current angle unit; these answers assume degrees (the start-up setting).*

### Example 5.9 — Common logarithm

Calculate **log 1000**. Press **Reset** first.

#1 SHIFT → #11 → #13 1 → #17 0 → #17 0 → #17 0 → #23 ) → #29 S⇔D

Display shows `3` — expected 3. ✔ verified on the app.

*log is base-10: it counts the zeros of a power of ten.*

### Example 5.10 — Natural logarithm

Calculate **ln 1**. Press **Reset** first.

#1 SHIFT → #15 → #13 1 → #23 ) → #29 S⇔D

Display shows `0` — expected 0. ✔ verified on the app.

*ln is the natural (base-e) logarithm.*

### Example 5.11 — Exponential $e^x$

Calculate **e^1**. Press **Reset** first.

#1 SHIFT → #16 → #13 1 → #23 ) → #29 S⇔D

Display shows `2.718281828` — expected 2.71828183. ✔ verified on the app.

*$e^x$ is the inverse of ln.*

### Example 5.12 — Scientific notation

**Calculate 6×10^2 × 3. Press Reset first.**

#11 6 → #1 SHIFT → #12 → #14 2 → #12 × → #15 3 → #29 S⇔D

**Display shows** `1800` **— expected 1800. ✔ verified on the app.**

*The exponent key enters ×10ⁿ without typing × 10 ^.*

### Example 5.13 — Area of a circle πr²

**Calculate π × 3². Press Reset first.**

#1 SHIFT → #9 → #12 × → #15 3 → #1 SHIFT → #13 → #29 S⇔D

**Display shows** `28.27433388` **— expected 28.27433388. ✔ verified on the app.**

*π is stored to full precision — never type 3.14.*

### Example 5.14 — Continue from the answer

**Calculate 100 × 4 = , then Ans ÷ 2. Press Reset first.**

#13 1 → #17 0 → #17 0 → #12 × → #9 4 → #29 S⇔D → #8 ÷ → #14 2 → #29 S⇔D

**Display shows** `200` **— expected 200. ✔ verified on the app.**

*Pressing an operator straight after a result continues from that answer (Ans).*

### Example 5.15 — Percent of an amount

**Calculate 15 % of 80. Press Reset first.**

#6 8 → #17 0 → #12 × → #13 1 → #10 5 → #1 SHIFT → #22 → #29 S⇔D

**Display shows** 12 **— expected 12. ✔ verified on the app.**

*The percent key turns the second number into a percentage of the first.*

### Example 5.16 — Average of three numbers

**Calculate (4 + 8 + 15) ÷ 3. Press Reset first.**

#22 ( → #9 4 → #20 + → #6 8 → #20 + → #13 1 → #10 5 → #23 ) → #8 ÷ → #15 3 → #29 S⇔D

**Display shows** 9 **— expected 9. ✔ verified on the app.**

*Add up the values, then divide by how many there are.*

### Example 5.17 — Mixed operations

**Calculate 50 − 4 × 6 + 7. Press Reset first.**

#10 5 → #17 0 → #16 − → #9 4 → #12 × → #11 6 → #20 + → #5 7 → #29 S⇔D

**Display shows** 33 **— expected 33. ✔ verified on the app.**

*Only the × part is done first; the − and + are then worked left to right.*

### Example 5.18 — Dividing twice

**Calculate 360 ÷ 4 ÷ 9. Press Reset first.**

#15 3 → #11 6 → #17 0 → #8 ÷ → #9 4 → #8 ÷ → #7 9 → #29 S⇔D

**Display shows** 10 **— expected 10. ✔ verified on the app.**

*Division is worked left to right: the first quotient is divided again.*

### Example 5.19 — Brackets inside a longer formula

**Calculate ((3 + 5) × 4 − 2) ÷ 10. Press Reset first.**

#22 ( → #22 ( → #15 3 → #20 + → #10 5 → #23 ) → #12 × → #9 4 → #16 − → #14 2 → #23 ) → #8 ÷ → #13 1 → #17 0 → #29 S⇔D

**Display shows** 3 **— expected 3. ✔ verified on the app.**

*The innermost brackets are worked out first, then the outer brackets, then the division.*

# 6 · Tricks & hacks

**Hack 1: Edit instead of retyping**

After a result, press ◀ or ▶ to bring the last expression back with the cursor in it; change one number and press = again.

**Hack 2: Ans chains**

Start a line with an operator (e.g. ×) and the calculator inserts Ans automatically — great for compound growth: 1.05 ×, then = = = …

**Hack 3: Exact vs decimal**

Natural-display models keep fractions, √ and π exact. Press S⇔D to flip to decimal; press it again to return.

**Hack 4: Close brackets for free**

Trailing closing brackets can be left off; = closes them for you.

**Hack 5: Implicit multiplication**

Type 2π or 3(4+1) without ×. Implicit multiplication binds tighter than ÷ on textbook models — use brackets if in doubt.

### Hack 6: Store constants

Store a value you reuse (e.g. g = 9.81) with STO and a letter; then type the letter in formulas.

### Hack 7: Power chains

x^y chains right to left: 2^3^2 = 2^9 = 512, not 64. Use brackets: (2^3)^2 = 64.

### Hack 8: Angles

Check the D / R / G indicator before trig. sin 30 = 0.5 only in degrees.

### Hack 9: Memory sums

M+ / M− keep a running total separate from the calculation line; RCL M shows it.

### Hack 10: Replay history

Use ▲ ▼ to scroll earlier lines; press = on an old line to re-run it.

### Hack 11: Hidden functions on this model

22 extra functions sit on shifted layers. Press the layer key once, then the key: for example #1 then #3 → INS; #1 then #4 → OFF; #1 then #5 → ; #1 then #6 → .

### Hack 12: Find any key by number

Use Appendix B (function finder): look up the function, read its key number, then find that number on the keypad. Numbers run left → right, top → bottom from #1 to #29.

### Hack 13: Type faster on a computer

21 keys on this model have a keyboard shortcut (see chapter 7). Typing digits and operators is much faster than clicking.

### Hack 14: Check before you trust

Estimate first (round each number) and compare with the display. If the answer is 10× or 100× off, a decimal point or percent key was missed.

### Hack 15: Copy results out

Press Copy in the bar above the calculator to put the display text on the clipboard instead of retyping it.

### Hack 16: Colour blind or low vision?

Hi-Contrast colours use black/white/yellow/blue/green/red with black or white text for maximum contrast; key numbers stay on in both colour schemes.

# 7 · Keyboard shortcuts

**With the calculator open on a computer, these physical keys press the numbered keys below (Ctrl/Alt/⌘ combinations are left to the browser). Shifted functions are reached by clicking the shift key first.**

| Keyboard | # | Key | Does |
|---|---|---|---|
| Backspace Delete | #3 | DEL▶ | DEL — delete the character at the cursor |
| 7 | #5 | 7 | Digit 7 |
| 8 | #6 | 8 | Digit 8 |
| 9 | #7 | 9 | Digit 9 |
| / | #8 | ÷ | Division ÷ |
| 4 | #9 | 4 | Digit 4 |
| 5 | #10 | 5 | Digit 5 |
| 6 | #11 | 6 | Digit 6 |
| * x | #12 | × | Multiplication × |
| 1 | #13 | 1 | Digit 1 |
| 2 | #14 | 2 | Digit 2 |
| 3 | #15 | 3 | Digit 3 |
| - | #16 | − | Subtraction − |
| 0 | #17 | 0 | Digit 0 |
| . , | #18 | • | Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored |
| r | #19 | √ | Square root √ |
| + | #20 | + | Addition + |
| ^ | #21 | x▭ | Power $x^y$ / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512) |
| ( | #22 | ( | Opening bracket ( |
| ) | #23 | ) | Closing bracket ). Closing brackets at the end of a line may be omitted |

| | | | |
|---|---|---|---|
| e | #24 | ×10x | ×10ˣ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10ˣ 23 |

## App shortcuts

| | |
|---|---|
| “# On / # Off” | show / hide key numbers |
| “Classic / Hi-Contrast” | switch key colours |
| “Copy” | copy display text |
| “Fit / 1:1” | scale to screen / design size |
| “Reset” | restart this calculator |
| Jump list | switch to another of the 45 calculators |

# 8 · Common mistakes

**Mistake 1: Using – instead of (–)**

The negative key (–) and the subtraction key are different; "– 5" at the start uses Ans – 5.

**Mistake 2: Wrong angle unit**

If sin 30 gives –0.988, the calculator is in radians. Switch to degrees in SETUP.

**Mistake 3: $-3^2$ vs $(-3)^2$**

$-3^2 = -9$ because the square is applied first. Type $(-3)^2$ to get 9.

**Mistake 4: Division precedence**

$1 \div 2\pi$ may be read as $1 \div (2\pi)$ on textbook models. Type $1 \div (2 \times \pi)$ explicitly to be sure.

**Mistake 5: Forgetting SHIFT is one-shot**

SHIFT applies only to the next key. If the S indicator is still lit, press SHIFT again to cancel.

**Mistake 6: Clearing memory by accident**

AC clears the line but not memory; CLR (SHIFT 9) can clear memory and variables.

### Mistake 7: Forgetting Reset between problems

Old values (memory, stack, open brackets, worksheet values) can leak into the next calculation. Press Reset or clear before starting a new problem.

### Mistake 8: Pressing the wrong layer

A shift / 2nd indicator that is still lit will turn the next key into its shifted function. Look at the display indicators before pressing.

# 9 · Practice drills

## Set A — Calculation drills (111)

**Work each one on calculator #41, then check your keys and the answer in Appendix A. Every answer was produced by this calculator.**

1. 25 + 5 × 6 = _____
2. 7 + 8 × 9 = _____
3. 150 + 12 × 11 = _____
4. 3 + 14 × 15 = _____
5. 48 + 6 × 7 = _____
6. 1000 + 25 × 4 = _____
7. 19 + 21 × 3 = _____
8. 64 + 8 × 8 = _____
9. (45 + 15) ÷ 6 = _____
10. (9 + 7) ÷ 2 = _____
11. (100 + 60) ÷ 8 = _____
12. (33 + 17) ÷ 5 = _____
13. (1 + 2) ÷ 4 = _____
14. (250 + 350) ÷ 12 = _____
15. (5 + 6) × (10 − 3) = _____
16. (12 + 3) × (8 − 5) = _____

17. (1.5 + 2.5) × (7 − 4) = _____

18. (20 + 30) × (15 − 5) = _____

19. (9 + 1) × (100 − 1) = _____

20. (11 + 13) × (17 − 9) = _____

21. $25^2$ = _____

22. $7^2$ = _____

23. $1.5^2$ = _____

24. $19^2$ = _____

25. $33^2$ = _____

26. $0.5^2$ = _____

27. $125^2$ = _____

28. $64^2$ = _____

29. $\sqrt{625}$ = _____

30. $\sqrt{2}$ = _____

31. $\sqrt{50}$ = _____

32. $\sqrt{1024}$ = _____

33. $\sqrt{3}$ = _____

34. $\sqrt{9801}$ = _____

35. 3^5 = _____

36. 1.05^3 = _____

37. 5^4 = _____

38. 7^3 = _____

39. 1.5^2 = _____

40. 10^6 = _____

41. 2^0.5 = _____

42. 1 ÷ 8 ($x^{-1}$) = _____

43. 1 ÷ 0.25 ($x^{-1}$) = _____

44. 1 ÷ 16 ($x^{-1}$) = _____

45. 1 ÷ 40 ($x^{-1}$) = _____

46. 1 ÷ 0.2 ($x^{-1}$) = _____

47. sin 90° (Deg mode) = _____

48. cos 0° (Deg mode) = _____

49. log 100 = _____

50. log 10000 = _____

51. ln 10 = _____

52. ln 2 = _____

53. e^2 = _____

54. 2×10^3 × 4 = _____

55. 1.5×10^4 × 2 = _____

56. 3×10^5 × 3 = _____

57. 4.5×10^2 × 2 = _____

58. π × $10^2$ = _____

59. π × $2.5^2$ = _____

60. π × $7^2$ = _____

61. π × $1.2^2$ = _____

62. 15 × 6 = , then Ans ÷ 9 = _____

63. 7.5 × 8 = , then Ans ÷ 3 = _____

64. 250 × 12 = , then Ans ÷ 100 = _____

65. 8 % of 250 = _____

66. 25 % of 360 = _____

67. 12.5 % of 40 = _____

68. 10 % of 450 = _____

69. 20 % of 125 = _____

70. 5 % of 640 = _____

71. 75 % of 48 = _____

72. 30 % of 90 = _____

73. (12 + 15 + 18) ÷ 3 = _____

74. (70 + 85 + 91) ÷ 3 = _____

75. (2.5 + 3.5 + 6) ÷ 3 = _____

76. (100 + 200 + 600) ÷ 3 = _____

77. 100 − 3 × 12 + 9 = _____

78. 18 − 2 × 5 + 40 = _____

79. 7.5 − 1.5 × 3 + 2 = _____

80. 64 − 8 × 4 + 10 = _____

81. 300 − 25 × 4 + 55 = _____

82. 9 − 3 × 3 + 1 = _____

83. 45 − 5 × 5 + 15 = _____

84. 1000 ÷ 8 ÷ 5 = _____

85. 96 ÷ 2 ÷ 3 = _____

86. 7200 ÷ 60 ÷ 60 = _____

87. 500 ÷ 4 ÷ 25 = _____

88. 81 ÷ 3 ÷ 3 = _____

89. ((10 + 2) × 3 − 6) ÷ 6 = _____

90. ((1.5 + 2.5) × 6 − 4) ÷ 5 = _____

91. ((20 + 30) × 2 − 25) ÷ 3 = _____

92. ((7 + 8) × 9 − 15) ÷ 12 = _____

93. 12 + 3 × 4 = _____

94. (12 + 8) ÷ 4 = _____

95. (3 + 4) × (9 − 2) = _____

96. $12^2$ = _____

97. $\sqrt{144}$ = _____

98. 2^10 = _____

99. 1 ÷ 4 ($x^{-1}$) = _____

100. tan 45° (Deg mode) = _____

101. log 1000 = _____

102. ln 1 = _____

103. e^1 = _____

104. 6×10^2 × 3 = _____

105. π × $3^2$ = _____

106. 100 × 4 = , then Ans ÷ 2 = _____

107. 15 % of 80 = _____

108. (4 + 8 + 15) ÷ 3 = _____

109. 50 − 4 × 6 + 7 = _____

110. 360 ÷ 4 ÷ 9 = _____

111. ((3 + 5) × 4 − 2) ÷ 10 = _____

## Set B — Key-finding drills (41)

**Without looking at the table, find each key on the picture. Answers in Appendix A.**

1. Which key is “SHIFT” (SHIFT / 2nd)? _____
2. Which key is “ALPHA” (ALPHA)? _____
3. Which key is “DEL▶” (DEL)? _____

4. **Which key is "AC/ON" (ON / AC)? _____**
5. **Which key is "÷" (Division ÷)? _____**
6. **Which key is "×" (Multiplication ×)? _____**
7. **Which key is "−" (Subtraction −)? _____**
8. **Which key is "•" (Decimal point. Pressing it first gives "0."; a second point in the same number is ignored)? _____**
9. **Which key is "√" (Square root √)? _____**
10. **Which key is "+" (Addition +)? _____**
11. **Which key is "x▭" (Power $x^y$ / ^)? _____**
12. **Which key is "(" (Opening bracket)? _____**
13. **Which key is ")" (Closing bracket ). Closing brackets at the end of a line may be omitted)? _____**
14. **Which key is "×10x" (×$10^x$ / EXP / EE)? _____**
15. **Which key is " ▲ " (Cursor ▲)? _____**
16. **Which key is " ▼ " (Cursor ▼)? _____**
17. **Which key is "◀" (Cursor ◀)? _____**
18. **Which key is "▶" (Cursor ▶)? _____**
19. **Which key is "S⇔D" (S⇔D)? _____**
20. **How do you reach "INS"? _____**
21. **How do you reach "OFF"? _____**
22. **How do you reach ""? _____**

**23. How do you reach “”? _____**

**24. How do you reach “”? _____**

**25. How do you reach “”? _____**

**26. How do you reach “”? _____**

**27. How do you reach “”? _____**

**28. How do you reach “”? _____**

**29. How do you reach “”? _____**

**30. How do you reach “”? _____**

**31. How do you reach “”? _____**

**32. How do you reach “”? _____**

**33. How do you reach “”? _____**

**34. How do you reach “”? _____**

**35. How do you reach “”? _____**

**36. How do you reach “”? _____**

**37. How do you reach “”? _____**

**38. How do you reach “”? _____**

**39. How do you reach “”? _____**

**40. How do you reach “”? _____**

**41. How do you reach “”? _____**

# Appendix A — Drill answers

## Set A

1. **55** (display `55`)

   `#14` 2 → `#10` 5 → `#20` + → `#10` 5 → `#12` × → `#11` 6 → `#29` S⇔D

2. **79** (display `79`)

   `#5` 7 → `#20` + → `#6` 8 → `#12` × → `#7` 9 → `#29` S⇔D

3. **282** (display `282`)

   `#13` 1 → `#10` 5 → `#17` 0 → `#20` + → `#13` 1 → `#14` 2 → `#12` × → `#13` 1 → `#13` 1 → `#29` S⇔D

4. **213** (display `213`)

   `#15` 3 → `#20` + → `#13` 1 → `#9` 4 → `#12` × → `#13` 1 → `#10` 5 → `#29` S⇔D

5. **90** (display `90`)

   `#9` 4 → `#6` 8 → `#20` + → `#11` 6 → `#12` × → `#5` 7 → `#29` S⇔D

6. **1100** (display `1100`)

   `#13` 1 → `#17` 0 → `#17` 0 → `#17` 0 → `#20` + → `#14` 2 → `#10` 5 → `#12` × → `#9` 4 → `#29` S⇔D

**7. 82** (display `82`)



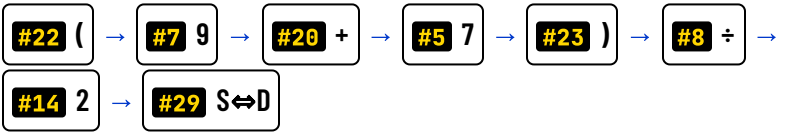

**8. 128** (display `128`)



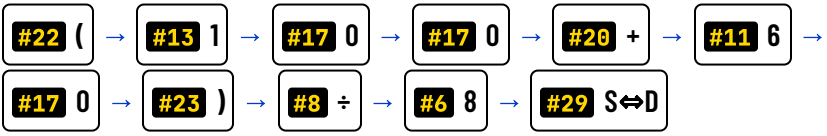

**9. 10** (display `10`)



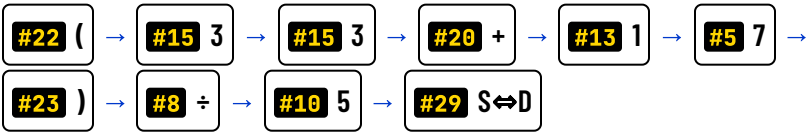

**10. 8** (display `8`)



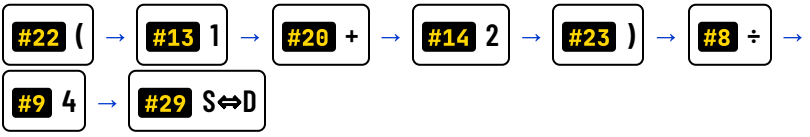

**11. 20** (display `20`)




**12. 10** (display `10`)




**13. 0.75** (display `0.75`)



### 14. 50 (display 50)

#22 ( → #14 2 → #10 5 → #17 0 → #20 + → #15 3 → #10 5 → #17 0 → #23 ) → #8 ÷ → #13 1 → #14 2 → #29 S⇔D

### 15. 77 (display 77)

#22 ( → #10 5 → #20 + → #11 6 → #23 ) → #12 × → #22 ( → #13 1 → #17 0 → #16 − → #15 3 → #23 ) → #29 S⇔D

### 16. 45 (display 45)

#22 ( → #13 1 → #14 2 → #20 + → #15 3 → #23 ) → #12 × → #22 ( → #6 8 → #16 − → #10 5 → #23 ) → #29 S⇔D

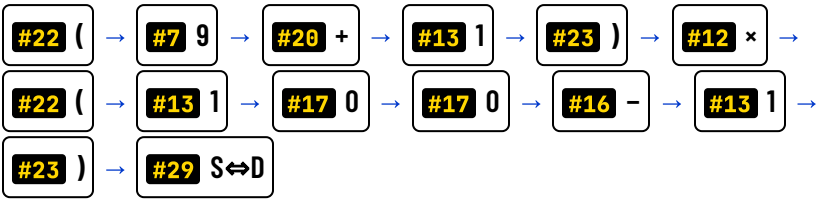
### 17. 12 (display 12)

#22 ( → #13 1 → #18 • → #10 5 → #20 + → #14 2 → #18 • → #10 5 → #23 ) → #12 × → #22 ( → #5 7 → #16 − → #9 4 → #23 ) → #29 S⇔D

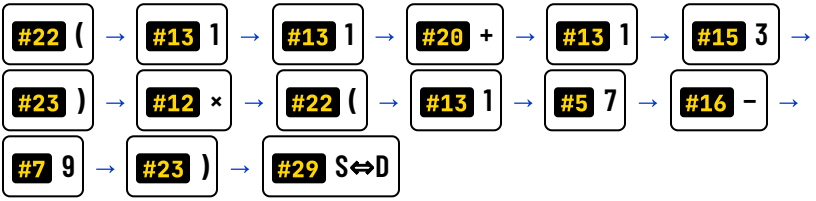
### 18. 500 (display 500)

#22 ( → #14 2 → #17 0 → #20 + → #15 3 → #17 0 → #23 ) → #12 × → #22 ( → #13 1 → #10 5 → #16 − → #10 5 → #23 ) → #29 S⇔D

19. **990** (display 990)



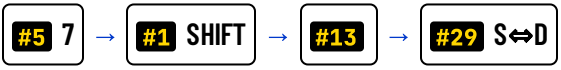

20. **192** (display 192)




21. **625** (display 625)




22. **49** (display 49)




23. **2.25** (display 2.25)




24. **361** (display 361)




25. **1089** (display 1089)




26. **0.25** (display 0.25)



**27. 15625** (display `15625`)




**28. 4096** (display `4096`)




**29. 25** (display `25`)




**30. 1.41421356** (display `1.414213562`)




**31. 7.07106781** (display `7.071067812`)




**32. 32** (display `32`)



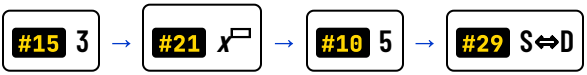

**33. 1.73205081** (display `1.732050808`)




**34. 99** (display `99`)




**35. 243** (display `243`)



**36. 1.157625** (display `1.157625`)

`#13` 1 → `#18` • → `#17` 0 → `#10` 5 → `#21` $x^{\square}$ → `#15` 3 → `#29` S⇔D

**37. 625** (display `625`)

`#10` 5 → `#21` $x^{\square}$ → `#9` 4 → `#29` S⇔D

**38. 343** (display `343`)

`#5` 7 → `#21` $x^{\square}$ → `#15` 3 → `#29` S⇔D

**39. 2.25** (display `2.25`)

`#13` 1 → `#18` • → `#10` 5 → `#21` $x^{\square}$ → `#14` 2 → `#29` S⇔D

**40. 1000000.000** (display `1000000`)

`#13` 1 → `#17` 0 → `#21` $x^{\square}$ → `#11` 6 → `#29` S⇔D

**41. 1.41421356** (display `1.414213562`)

`#14` 2 → `#21` $x^{\square}$ → `#17` 0 → `#18` • → `#10` 5 → `#29` S⇔D

**42. 0.125** (display `0.125`)

`#6` 8 → `#1` SHIFT → `#14` → `#29` S⇔D

**43. 4** (display `4`)

`#17` 0 → `#18` • → `#14` 2 → `#10` 5 → `#1` SHIFT → `#14` → `#29` S⇔D

**44. 0.0625** (display `0.0625`)

`#13` 1 → `#11` 6 → `#1` SHIFT → `#14` → `#29` S⇔D

**45. 0.025** (display `0.025`)

`#9` 4 → `#17` 0 → `#1` SHIFT → `#14` → `#29` S⇔D

**46. 5** (display `5`)

`#17` 0 → `#18` • → `#14` 2 → `#1` SHIFT → `#14` → `#29` S⇔D

**47. 1** (display `1`)

`#1` SHIFT → `#6` → `#7` 9 → `#17` 0 → `#23` ) → `#29` S⇔D

**48. 1** (display `1`)

`#1` SHIFT → `#7` → `#17` 0 → `#23` ) → `#29` S⇔D

**49. 2** (display `2`)

`#1` SHIFT → `#11` → `#13` 1 → `#17` 0 → `#17` 0 → `#23` ) → `#29` S⇔D

**50. 4** (display `4`)

`#1` SHIFT → `#11` → `#13` 1 → `#17` 0 → `#17` 0 → `#17` 0 → `#17` 0 → `#23` ) → `#29` S⇔D

**51. 2.30258509** (display `2.302585093`)

`#1` SHIFT → `#15` → `#13` 1 → `#17` 0 → `#23` ) → `#29` S⇔D

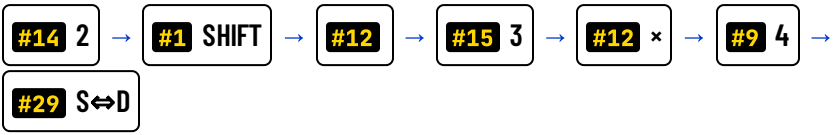
**52. 0.69314718** (display `0.6931471806`)

`#1` SHIFT → `#15` → `#14` 2 → `#23` ) → `#29` S⇔D

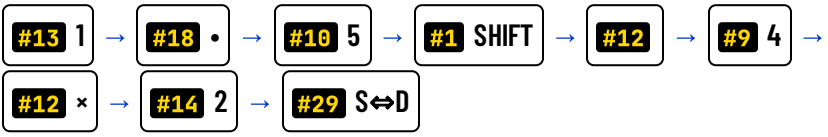
**53. 7.3890561** (display `7.389056099`)

`#1` SHIFT → `#16` → `#14` 2 → `#23` ) → `#29` S⇔D

**54. 8000** (display `8000`)



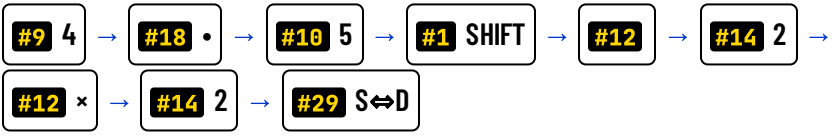

**55. 30000** (display `30000`)



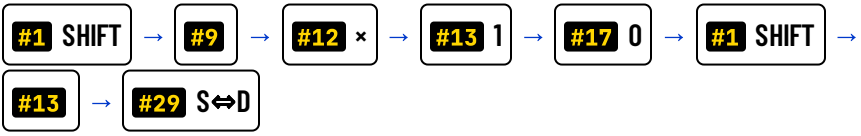

**56. 900000** (display `900000`)



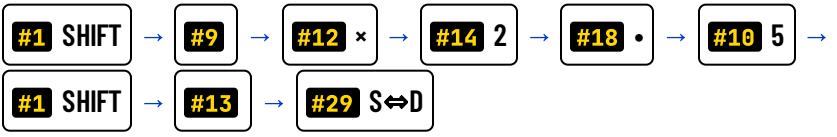

**57. 900** (display `900`)




**58. 314.15926536** (display `314.1592654`)




**59. 19.63495408** (display `19.63495408`)




**60. 153.93804003** (display `153.93804`)



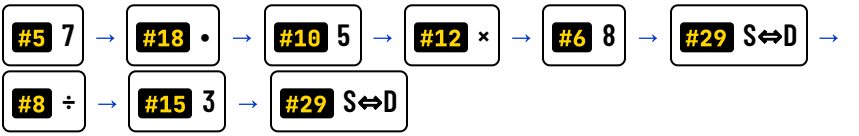
**61. 4.52389342** (display `4.523893421`)



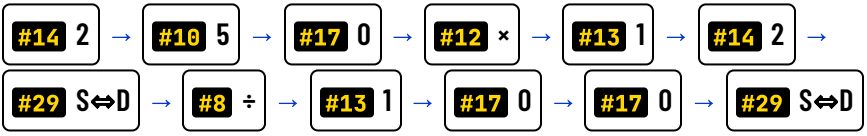

**62. 10** (display `10`)




**63. 20** (display `20`)



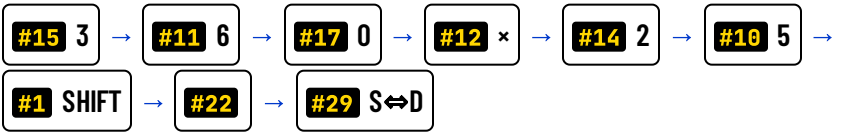

**64. 30** (display `30`)



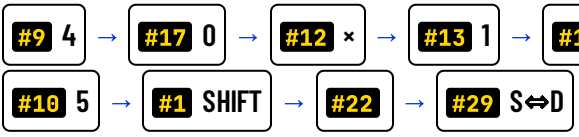

**65. 20** (display `20`)




**66. 90** (display `90`)




**67. 5** (display `5`)



**68. 45** (display `45`)

`#9` 4 → `#10` 5 → `#17` 0 → `#12` × → `#13` 1 → `#17` 0 → `#1` SHIFT → `#22` → `#29` S⇔D

**69. 25** (display `25`)

`#13` 1 → `#14` 2 → `#10` 5 → `#12` × → `#14` 2 → `#17` 0 → `#1` SHIFT → `#22` → `#29` S⇔D

**70. 32** (display `32`)

`#11` 6 → `#9` 4 → `#17` 0 → `#12` × → `#10` 5 → `#1` SHIFT → `#22` → `#29` S⇔D

**71. 36** (display `36`)

`#9` 4 → `#6` 8 → `#12` × → `#5` 7 → `#10` 5 → `#1` SHIFT → `#22` → `#29` S⇔D

**72. 27** (display `27`)

`#7` 9 → `#17` 0 → `#12` × → `#15` 3 → `#17` 0 → `#1` SHIFT → `#22` → `#29` S⇔D

**73. 15** (display `15`)

`#22` ( → `#13` 1 → `#14` 2 → `#20` + → `#13` 1 → `#10` 5 → `#20` + → `#13` 1 → `#6` 8 → `#23` ) → `#8` ÷ → `#15` 3 → `#29` S⇔D

**74. 82** (display 82)

#22 ( → #5 7 → #17 0 → #20 + → #6 8 → #10 5 → #20 + → #7 9 → #13 1 → #23 ) → #8 ÷ → #15 3 → #29 S⇔D

**75. 4** (display 4)

#22 ( → #14 2 → #18 • → #10 5 → #20 + → #15 3 → #18 • → #10 5 → #20 + → #11 6 → #23 ) → #8 ÷ → #15 3 → #29 S⇔D

**76. 300** (display 300)

#22 ( → #13 1 → #17 0 → #17 0 → #20 + → #14 2 → #17 0 → #17 0 → #20 + → #11 6 → #17 0 → #17 0 → #23 ) → #8 ÷ → #15 3 → #29 S⇔D

**77. 73** (display 73)

#13 1 → #17 0 → #17 0 → #16 − → #15 3 → #12 × → #13 1 → #14 2 → #20 + → #7 9 → #29 S⇔D

**78. 48** (display 48)

#13 1 → #6 8 → #16 − → #14 2 → #12 × → #10 5 → #20 + → #9 4 → #17 0 → #29 S⇔D

**79. 5** (display 5)

#5 7 → #18 • → #10 5 → #16 − → #13 1 → #18 • → #10 5 → #12 × → #15 3 → #20 + → #14 2 → #29 S⇔D

**80. 42 (display 42)**

#11 6 → #9 4 → #16 − → #6 8 → #12 × → #9 4 → #20 + → #13 1 → #17 0 → #29 S⇔D

**81. 255 (display 255)**

#15 3 → #17 0 → #17 0 → #16 − → #14 2 → #10 5 → #12 × → #9 4 → #20 + → #10 5 → #10 5 → #29 S⇔D

**82. 1 (display 1)**

#7 9 → #16 − → #15 3 → #12 × → #15 3 → #20 + → #13 1 → #29 S⇔D

**83. 35 (display 35)**

#9 4 → #10 5 → #16 − → #10 5 → #12 × → #10 5 → #20 + → #13 1 → #10 5 → #29 S⇔D

**84. 25 (display 25)**

#13 1 → #17 0 → #17 0 → #17 0 → #8 ÷ → #6 8 → #8 ÷ → #10 5 → #29 S⇔D

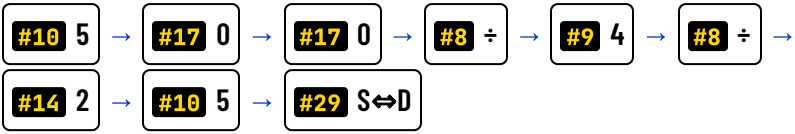
**85. 16 (display 16)**

#7 9 → #11 6 → #8 ÷ → #14 2 → #8 ÷ → #15 3 → #29 S⇔D

**86. 2 (display 2)**

#5 7 → #14 2 → #17 0 → #17 0 → #8 ÷ → #11 6 → #17 0 → #8 ÷ → #11 6 → #17 0 → #29 S⇔D

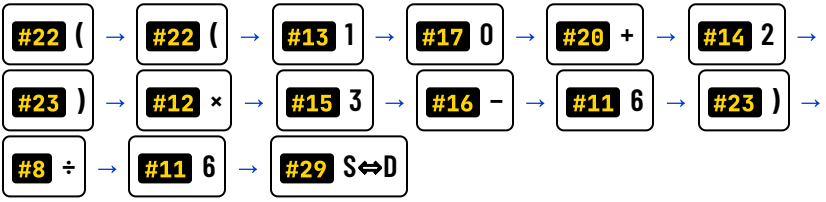
**87. 5** (display `5`)



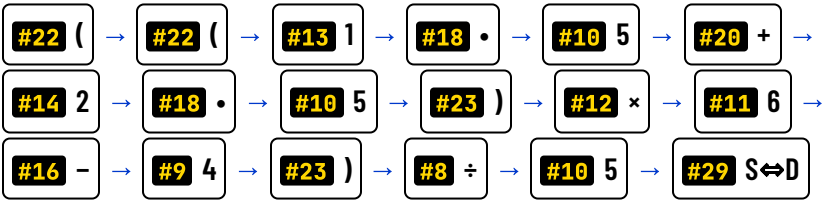

**88. 9** (display `9`)



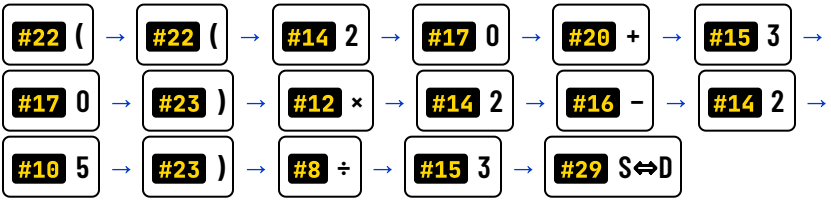

**89. 5** (display `5`)



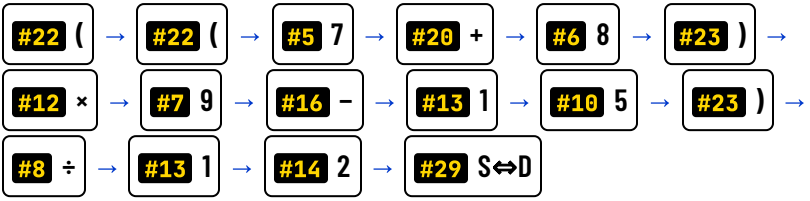

**90. 4** (display `4`)




**91. 25** (display `25`)



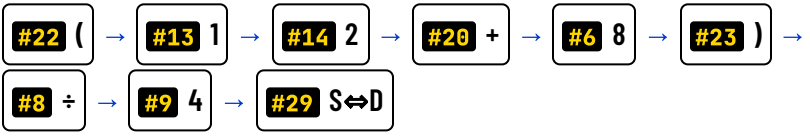
**92. 10** (display `10`)



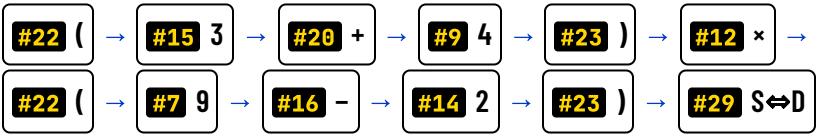

**93. 24** (display `24`)




**94. 5** (display `5`)




**95. 49** (display `49`)




**96. 144** (display `144`)



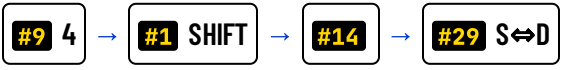

**97. 12** (display `12`)




**98. 1024** (display `1024`)




**99. 0.25** (display `0.25`)



**100. 1** (display `1`)




**101. 3** (display `3`)




**102. 0** (display `0`)




**103. 2.71828183** (display `2.718281828`)



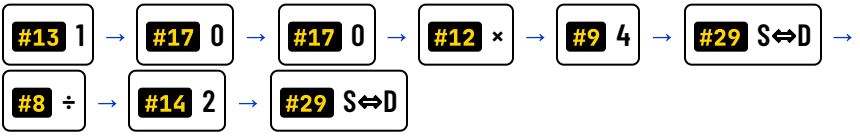

**104. 1800** (display `1800`)




**105. 28.27433388** (display `28.27433388`)




**106. 200** (display `200`)



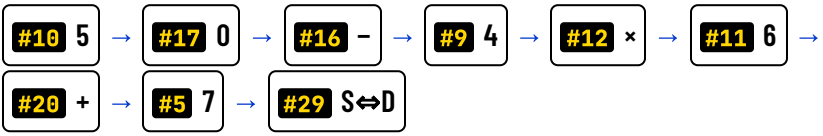

**107. 12** (display `12`)



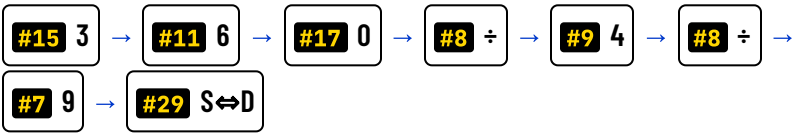
### 108. 9 (display 9)



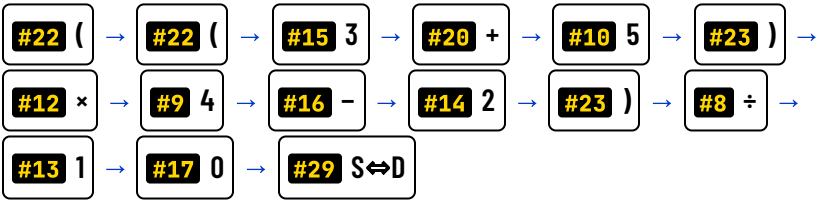

### 109. 33 (display 33)




### 110. 10 (display 10)




### 111. 3 (display 3)




## Set B

1. #1
2. #2
3. #3
4. #4
5. #8
6. #12
7. #16
8. #18
9. #19
10. #20
11. #21
12. #22
13. #23
14. #24
15. #25
16. #26
17. #27
18. #28
19. #29
20. #1 SHIFT then #3 DEL▶
21. #1 SHIFT then #4 AC/ON
22. #1 SHIFT then #5 7
23. #1 SHIFT then #6 8
24. #1 SHIFT then #7 9
25. #1 SHIFT then #8 ÷
26. #1 SHIFT then #9 4

27. #1 SHIFT then #10 5
28. #1 SHIFT then #11 6
29. #1 SHIFT then #12 ×
30. #1 SHIFT then #13 1
31. #1 SHIFT then #14 2
32. #1 SHIFT then #15 3
33. #1 SHIFT then #16 -
34. #1 SHIFT then #17 0
35. #1 SHIFT then #18 •
36. #1 SHIFT then #19 √
37. #1 SHIFT then #21 x▭
38. #1 SHIFT then #22 (
39. #1 SHIFT then #23 )
40. #1 SHIFT then #24 ×10x
41. #1 SHIFT then #29 S⇔D

# Appendix B — Function finder (A–Z)

| Function | Label | Keys |
|---|---|---|
| ×10^x / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10^x 23 | exp10 | #1 → #12 |
| ×10^x / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10^x 23 | ×10x | #24 |
| Addition + | + | #20 |
| ALPHA — the next key types its letter/variable (red/green label) | ALPHA | #2 |
| Ans — recall the last answer into the current expression | ans | #1 → #24 |
| Closing bracket ). Closing brackets at the end of a line may be omitted | ) | #23 |
| Comma / argument separator (between function arguments, list items or coordinates) | comma | #1 → #23 |
| Common logarithm $\log_{10}$ | log | #1 → #11 |
| Cosine (current angle unit) | cos | #1 → #7 |
| Cube root $\sqrt[3]{}$ | cbrt | #1 → #19 |
| Cube $x^3$ | cube | #1 → #17 |
| Cursor ▲ — up / previous history line / previous item | ▲ | #25 |
| Cursor ▶ — move right in the line | ▶ | #28 |
| Cursor ▼ — down / next history line / next item | ▼ | #26 |
| Cursor ◀ — move left in the line | ◀ | #27 |
| Decimal point. Pressing it first gives "0."; a second point in the same number is ignored | • | #18 |
| Degrees-minutes-seconds entry/display °′ ″ | dms | #1 → #8 |
| DEL — delete the character at the cursor | DEL▶ | #3 |
| Digit 0 | 0 | #17 |
| Digit 1 | 1 | #13 |

| | | |
|---|---|---|
| **Digit 2** | **2** | **#14** |
| **Digit 3** | **3** | **#15** |
| **Digit 4** | **4** | **#9** |
| **Digit 5** | **5** | **#10** |
| **Digit 6** | **6** | **#11** |
| **Digit 7** | **7** | **#5** |
| **Digit 8** | **8** | **#6** |
| **Digit 9** | **9** | **#7** |
| **Division ÷** | **÷** | **#8** |
| **$e^x$ — exponential** | **epow** | **#1 → #16** |
| **Hyperbolic prefix — the next sin/cos/tan becomes sinh/cosh/tanh** | **hyp** | **#1 → #5** |
| **INS — insert mode at the cursor** | **INS** | **#1 → #3** |
| **M+ — add the value to independent memory M** | **mplus** | **#1 → #29** |
| **Multiplication ×** | **×** | **#12** |
| **n-th root $\sqrt[x]{y}$** | **nroot** | **#1 → #21** |
| **Natural logarithm ln** | **ln** | **#1 → #15** |
| **OFF — power off (state is lost; press ON to restart)** | **OFF** | **#1 → #4** |
| **ON / AC — power on, or clear all** | **AC/ON** | **#4** |
| **Opening bracket (** | **(** | **#22** |
| **Percent %** | **pct** | **#1 → #22** |
| **Power $x^y$ / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)** | **x▭** | **#21** |
| **Random number 0–1** | **ran** | **#1 → #18** |
| **Reciprocal $x^{-1}$ (1/x)** | **inv** | **#1 → #14** |
| **S⇔D — toggle the result between exact (fraction, √, π) and decimal** | **S⇔D** | **#29** |

| | | |
|---|---|---|
| SHIFT / 2nd — the next key uses its upper (shift-coloured) function | SHIFT | #1 |
| Sine (uses the current angle unit) | sin | #1 → #6 |
| Square root √ | √ | #19 |
| Square $x^2$ | sq | #1 → #13 |
| Subtraction – | – | #16 |
| Tangent (current angle unit) | tan | #1 → #10 |
| π (3.14159265…) as an exact constant | pi | #1 → #9 |

# Appendix C — Glossary

**Ans**
The result of the last calculation, reusable in the next.

**AOS**
Algebraic Operating System — operator precedence on a one-line display.

**RPN**
Reverse Polish Notation — operands first, then the operation; no = key.

**Stack**
The RPN registers X, Y, Z, T where numbers wait for operations.

**TVM**
Time Value of Money — n, i, PV, PMT, FV.

**Natural display**
Shows fractions, roots and powers as printed in textbooks.

**Shift / 2nd**
Prefix key that selects the function printed above a key.

**Alpha**
Prefix key that types the letter printed beside a key.

**Radian**
Angle unit; 2π rad = 360°.

**Grad**
Angle unit; 400 grad = 360°.

**Engineering notation**
Scientific notation with exponents that are multiples of 3.

**Mantissa**
The digits of a number in scientific notation.

**Memory (M)**

An independent register for running totals.

**Mark-up**

Increase over cost; margin is profit as a percent of selling price.

**Grand total (GT)**

Sum of all totals since the last clear.

**nCr / nPr**

Combinations / permutations of n items taken r at a time.

**Factorial**

n! = 1×2×…×n.

**Reciprocal**

1/x.

**Standard deviation**

Spread of data around the mean (σ population, s sample).

**Key number**

The small badge 1…N on each key in U-45 CALCULATORS, used throughout this manual.

# Appendix D — Model notes & limits

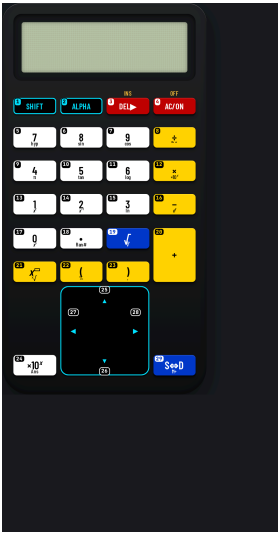
Pocket model: functions printed under each key are reached with SHIFT. The photo shows no "=" key, so S⇔D works as "=" (and toggles fraction/decimal once a result is shown); Enter on the keyboard also evaluates. Some garbled key faces ($0^3$, $\sqrt{}^3$, ( ),1) were mapped to $x^y$, ( and ).

## Planned (not yet working)

- Statistics mode
- Base-N mode
- Equation solver
- Table mode
- Constants
- Unit conversions
- Replay history
- Complex numbers
- Exam mode (locks memory & shows indicator)
- Answer history export to text
- Speech read-out of results
- Colour-blind safe key theme
- Custom shortcut keys

U-45 CALCULATORS models are independent software recreations; they are not affiliated with any calculator manufacturer and carry no third-party trademarks.

# Appendix E — Quick-reference card




| | | |
|---|---|---|
| #1 SHIFT | #2 ALPHA | #3 DEL▶ ⇧ INS |
| #4 AC/ON ⇧ OFF | #5 7 | #6 8 |
| #7 9 | #8 ÷ | #9 4 |
| #10 5 | #11 6 | #12 × |
| #13 1 | #14 2 | #15 3 |
| #16 − | #17 0 | #18 • |
| #19 √ | #20 + | #21 x▭ |
| #22 ( | #23 ) | #24 ×10x |
| #25 ▲ | #26 ▼ | #27 ◀ |
| #28 ▶ | #29 S⇔D | |

U-45 CALCULATORS · Calculator #41 · cut out and keep